

Experiment-15

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler
data = {
    "Name": ["Arun", "Bala", "Charan", "Divya", "Ezhil", "Bala", "Farah", "Gokul"],
    "Age": [20, 21, np.nan, 22, -5, 21, 20, 23],
    "Department": ["CSE", "cse", "ECE", "ECE", "CSE", "cse", "IT", "IT"],
    "Marks": [85, 90, 75, np.nan, 120, 90, 65, 80]
}
df = pd.DataFrame(data)
print(df)
```

	Name	Age	Department	Marks
0	Arun	20.0	CSE	85.0
1	Bala	21.0	cse	90.0
2	Charan	NaN	ECE	75.0
3	Divya	22.0	ECE	NaN
4	Ezhil	-5.0	CSE	120.0
5	Bala	21.0	cse	90.0
6	Farah	20.0	IT	65.0
7	Gokul	23.0	IT	80.0

```
print("Missing Values:")
print(df.isnull().sum())
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Marks"] = df["Marks"].fillna(df["Marks"].median())
print("\nAfter Handling Missing Values:")
print(df)
```

Missing Values:

```
Name      0
Age        1
Department 0
Marks      1
dtype: int64
```

After Handling Missing Values:

	Name	Age	Department	Marks
0	Arun	20.0	CSE	85.0
1	Bala	21.0	cse	90.0
2	Charan	21.0	ECE	75.0
3	Divya	22.0	ECE	85.0
4	Ezhil	-5.0	CSE	120.0
5	Bala	21.0	cse	90.0
6	Farah	20.0	IT	65.0
7	Gokul	23.0	IT	80.0

```
df.loc[df["Age"] < 0, "Age"] = np.nan
df.loc[(df["Marks"] < 0) | (df["Marks"] > 100), "Marks"] = np.nan
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Marks"] = df["Marks"].fillna(df["Marks"].median())
print(df)
```

	Name	Age	Department	Marks
0	Arun	20.0	CSE	85.0
1	Bala	21.0	cse	90.0
2	Charan	21.0	ECE	75.0
3	Divya	22.0	ECE	85.0
4	Ezhil	21.0	CSE	85.0
5	Bala	21.0	cse	90.0
6	Farah	20.0	IT	65.0
7	Gokul	23.0	IT	80.0

```
df = df.drop_duplicates()
print("Dataset After Removing Duplicates:")
print(df)
```

Dataset After Removing Duplicates:

	Name	Age	Department	Marks
0	Arun	20.0	CSE	85.0
1	Bala	21.0	cse	90.0
2	Charan	21.0	ECE	75.0
3	Divya	22.0	ECE	85.0
4	Ezhil	21.0	CSE	85.0
6	Farah	20.0	IT	65.0
7	Gokul	23.0	IT	80.0

```
df["Department"] = (df["Department"].str.strip().str.upper())  
print(df["Department"])
```

```
0    CSE  
1    CSE  
2    ECE  
3    ECE  
4    CSE  
6     IT  
7     IT  
Name: Department, dtype: object
```